RISHIT TANDON

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PROFESSIONAL SUMMARY

Final-year Computer Science student (B.Tech, SRM IST, 2027) who ships real products across agentic AI, full-stack web, and embedded systems. Built **7 live products**, won **12+ hackathons** (6 first-place finishes), and maintain **100+ public GitHub repositories**. Hands-on with LLM agents, RAG pipelines, payment integrations, and production deployments.

EDUCATION

B.Tech, Computer Science & Engineering | SRM Institute of Science and Technology *2023 – 2027 (Expected)*
CGPA: 8.05 (Last Semester: 9.3) | Faculty-selected: Metaverse (Top 80/batch), Responsible AI (Top 120/batch).

Senior Secondary (XII), CBSE Science | Delhi Public School, Lucknow *2023*

Secondary (X), ICSE | Seth M. R. Jaipuria School, Lucknow *2021*

TECHNICAL SKILLS

Languages: Python, Java, C++, JavaScript, SQL, C
AI/ML: PyTorch, TensorFlow, Scikit-learn, OpenCV, Hugging Face, LangChain, LangGraph, RAG, Agentic AI
Web & Backend: React.js, Next.js, Tailwind CSS, FastAPI, Flask, REST APIs, Webhooks, OAuth 2.0, PWA
Databases: PostgreSQL, MySQL, Supabase, Neo4j (Graph DB), ChromaDB (Vector DB)
DevOps & Cloud: Git, GitHub Actions, Docker, CI/CD, AWS, Vercel, Linux
Embedded & IoT: ESP32, ESP32-CAM, Arduino, Motor Drivers, IoT Digital Twins, Unity

PROFESSIONAL EXPERIENCE

Research Intern – ML & Image Processing | SRM IST in collaboration with NIDM *Oct 2025 – Present*

- Built an automated GLOF (glacial lake outburst flood) early-warning pipeline using Python and Earth Engine.
- Processed 35+ years of multimodal satellite imagery (Sentinel-1 SAR, Sentinel-2, Landsat-5) with Rasterio.
- Engineered SAR and multispectral features (NDWI, NDSI) for a Decision Tree, cutting false positives by ~18%.

Software Development Intern | DGTL Innovations *May 2025 – Jul 2025*

- Built REST APIs with Python and MySQL; optimized queries and indexing to cut average response time by ~20%.
- Developed responsive frontend modules in HTML, CSS, and JavaScript for production pages.
- Worked in a Git-based pull request and code review workflow alongside the core development team.

KEY PROJECTS

AgentLab – AI Agent Marketplace | GitHub | Live | *Next.js, FastAPI, Supabase, Razorpay, AWS*

- Built and deployed a marketplace where developers buy and integrate AI agents via auto-provisioned REST APIs.
- Implemented multi-tenant authentication, API key management, and request isolation on FastAPI + Supabase.
- Automated payments and provisioning with Razorpay webhooks for real-time API key generation and delivery.

SplitsFare – Group Expense Splitting App | Live | *React, Vite, Supabase, Razorpay, UPI, PWA*

- Shipped an installable PWA for groups to split expenses and settle instantly through UPI deep links.
- Secured data with PostgreSQL row-level security and Google OAuth; migrated auth from Firebase to Supabase.

LinkedOut – AI LinkedIn Content Platform | Live | *Next.js, FastAPI, Supabase, Gemini API, LinkedIn OAuth*

- Launched a Gemini-powered PWA that writes LinkedIn posts in each user's voice using persona profiles.
- Integrated LinkedIn OAuth 2.0 and the LinkedIn API for scheduled auto-publishing on FastAPI + Supabase.
- Built an analytics dashboard to track post engagement and performance over time.

DocMind – Agentic Graph RAG Platform | *Python, LangGraph, LangChain, Qwen3, Neo4j, ChromaDB*

- Built an agentic Graph RAG platform for PDF ingestion, semantic retrieval, and LLM-powered question answering using Neo4j knowledge graphs and ChromaDB vector search.
- Developed multi-agent workflows with LangGraph, incorporating memory, personas, and tool-calling to enhance document understanding and contextual reasoning.
- Designed an evaluation framework using retrieval metrics, semantic similarity, and keyword-based scoring to measure response quality and reduce hallucinations.

QML·PLACE – Quantum ML VLSI Placement Engine | GitHub | *Python, PennyLane, PyTorch, OpenROAD, Docker*

- VQC-based placer as a drop-in for OpenROAD RePlAce: 8–15% HPWL gain, 1.31× speedup on Sky130 HD.
- Containerized the EDA flow with Docker + CUDA, with a PyQt6 GUI for live HPWL and GPU tracking.

AI Smart Glasses (AISGR) | *Python, ESP32-CAM, LLaVA, Llama 3.2 Vision, FastAPI*

- Built a wearable AI assistant powered by Vision-Language Models for real-time scene understanding, object recognition, visual question answering, and voice interaction.
- Architected a multi-agent system integrating vision, memory, and planning agents with local Ollama models and OpenAI fallbacks for multimodal reasoning.
- Developed ESP32-CAM firmware with gesture controls, MJPEG streaming, and real-time edge-to-cloud communication for assistive applications.

NeuroVibe – Brain-Controlled Smart Wheelchair | GitHub | *ESP32, BioAmp EXG, Scikit-learn, Arduino*

- Built a Brain-Computer Interface (BCI) system using EEG signals (BioAmp Band + ESP32) to control wheelchair movement in real time.
- Trained a Random Forest classifier to decode mental commands (Move, Stop, LED ON/OFF) from processed EEG data.
- Delivered a low-latency, low-cost assistive mobility solution integrating the ML model with motor control and appliance systems.

AI Negotiation System | GitHub | *Python, Flask, Gemini 2.0, Playwright, Llama 3.2*

- Built a buyer-vs-seller multi-agent negotiation system on Gemini 2.0 with recursive feedback loops.
- Added a Llama 3.2 LLM-as-judge evaluator and a Playwright scraper with Google Shopping fallback.

Multiverse – IoT Digital Twin Platform | Demo | *Next.js, Unity, C#, ESP32, REST APIs*

- Built a real-time Unity 3D digital twin syncing live telemetry from 20+ ESP nodes (100+ sensor data points).
- Implemented bi-directional remote actuation via a custom API, improving monitoring responsiveness by 40%.

More live products: SRM AutoPilot | GeoTag Pro | Git Maxer | Aveum

HACKATHONS & AWARDS

- **1st Place:** RoboRoarZ Singapore 2024 (SUTD/IEEE), SEISMO HACK 1.0 (National), SRM ML Project Expo, South India Conclave, Semiconductor Expo, NWC Expo.
- **2nd Place:** RoboRoarZ International 2024 (James Dyson Foundation), Arduino Hackathon, C.Tech Project Expo.
- **3rd Place & Special Mentions:** BIS Hackathon (3rd), Hack Sustain, SRM Founder's Hackathon.

LEADERSHIP

- Co-organized International RoboFest 2.0 at SRM IST with 68+ competing teams; built the judging portal.

CERTIFICATIONS

- Microsoft Azure Essentials – Microsoft & Great Learning (2026)
- Machine Learning – NPTEL (IIT Kharagpur)
- Python for Data Science, AI & Development – IBM
- Getting Started with AI on Jetson Nano – NVIDIA